Ankit Patidar

+91-7489483739 | patidarankit3739@gmail.com | linkedin.com/in/ankitpatidar | github.com/ankitpatidar3739

Education

Vellore Institute of Technology Bhopal Sep 2023 – Present
B.Tech, CSE (AI & ML) | CGPA: 8.68 / 10

N.S. Singhvi Higher Secondary School, Mandsaur Apr 2022
HSC – 83.8%

Shri Sai Public School, Dalauda Apr 2020
SSC – 80.2%

Technical Skills

Languages: Python, C++
ML / DL: Regression, Classification, ANN, CNN, RNN, LSTM, Transformers, Fine-tuning LLMs
GenAI & LLM Stack: LangChain, RAG, Vector Databases, Prompt Engineering, LLM APIs (Groq / OpenAI / Gemini)
API Development: FastAPI, REST APIs, Pydantic, Uvicorn
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV
Tools & Platforms: VS Code, Git, Google Colab, Jupyter Notebook, Docker

Experience

Research Intern — INCOIS, Hyderabad May 2026 – Jun 2026
Ministry of Earth Sciences, Govt. of India

- Built an end-to-end marine macro-litter detection pipeline on EOS-06 OCM (OCEANSAT-3) imagery, processing 13 spectral bands across large-scale coastal zones.
- Engineered 78 spectral indices (FDI, NDVI, PI) with adaptive thresholding and multi-date persistence mapping, reducing atmospheric false positives by ~40%.
- Delivered 5+ production-ready GIS outputs — GeoTIFF layers, NetCDF files, and diagnostic composites — for coastal debris surveillance and prioritisation.

Projects

Marine Macro-Litter Detection — EOS-06 OCM Imagery Python, Remote Sensing, OpenCV, AI
Research Project — INCOIS, Ministry of Earth Sciences, Govt. of India

- Processed 13 spectral bands of OCEANSAT-3 data, computing 78 normalised difference indices (FDI, NDVI, PI) with adaptive DN-scale thresholding to detect marine plastic debris across coastal zones.
- Applied multi-date persistence mapping across 3+ satellite passes, suppressing ~40% atmospheric false positives; generated GeoTIFF and NetCDF outputs integrated with GIS tools for surveillance reporting.
- Produced 5+ georeferenced deliverables — diagnostic PNGs, composite maps, and GeoTIFF layers — enabling targeted debris collection with significantly reduced manual inspection overhead.

DocuChat — RAG-Powered PDF Chatbot Python, LangChain, FAISS, FastAPI, Streamlit
Personal Project | github.com/ankitpatidar3739

- Built a conversational RAG pipeline over PDFs using LangChain LCEL, FAISS vector store, and Groq (LLaMA 3.1-8B) with HuggingFace embeddings; page-level source citations on 100% of answers.
- Engineered a 2-stage condense + retrieve + generate pipeline for multi-turn conversation, reducing retrieval failure on follow-up queries to 0% via standalone question condensation before FAISS lookup.
- Exposed 4 REST endpoints via FastAPI with Pydantic models, file upload validation, and auto-generated Swagger docs; Streamlit UI and FastAPI share all src/ modules with 0 code duplication.

Certifications

- Applied Machine Learning in Python — Coursera
- Google Cloud Generative AI — SmartInternz
- Google IT Support Certificate — Google / Credly

Leadership & Achievements

- Management Lead, Music and Band Club — VIT Bhopal
- Football Runner-Up, Sports Fest AdVITya’24 — VIT Bhopal